

Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

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Population closure and equilibrium transition

Previous focus: the population closure linking fertility to housing demand:

births today $\longrightarrow$ new households later $\longrightarrow$ housing demand and prices.

This presentation embeds the closure in equilibrium:

- a simplified economy with a well-defined steady state;
- the impact response and subsequent demographic adjustment;
- the same population law in the lifecycle quantitative model;
- calibration and validation from 2007 to 2023, followed by a forward transition.

A toy economy

To fix ideas, consider a simplified economy with young and old households:

- Y_t is the number of young households at date t ; O_t is the number of old households.
- Total adult-household population is $S_t = Y_t + O_t$.
- Young households choose consumption, housing services, and n_t **household children**; next period they are old.
- Each household child becomes one young household in the next generation.
- Both generations purchase housing services in the same market at price p_t .

The cohort law is therefore

$$Y_{t+1} = n_t Y_t, \quad O_{t+1} = Y_t,$$

so $n_t = 1$ is replacement.

The quantitative model adds tenure choice, mortgages, saving, and bequests.

Household choice and housing-market clearing

A young household earns y and chooses consumption c , housing services h , and household children n :

$$u(c, h, n) = c + \alpha \log h + \nu \log n, \quad c + ph + [q + a(p + \omega)]n = y.$$

Here α values adult housing, ν values children, and each household child requires a units of housing plus q other goods. The term ω is the extra cost of accessing child-suitable housing.

The interior choices are

$$n(p) = \frac{\nu}{q + a(p + \omega)}, \quad h_Y(p) = \frac{\alpha}{p} + an(p).$$

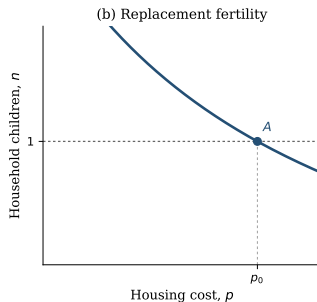
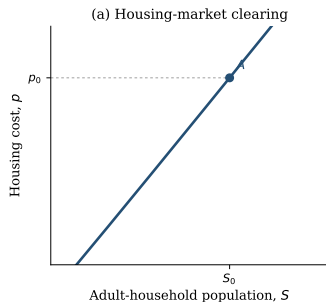
An old household has utility $c_O + \bar{h}_O \log h_O$, so its housing demand is $h_O(p) = \bar{h}_O/p$. Housing supply and market clearing are

$$H^s(p) = \bar{H}p^\varepsilon, \quad Y_t h_Y(p_t) + O_t h_O(p_t) = H^s(p_t).$$

The parameter $\bar{H}$ scales housing supply and ε is its price elasticity.

Higher housing costs reduce housing demand and fertility; cohort sizes feed back into the market-clearing price.

Initial steady state



A steady state is a constant triple (Y^*, O^*, p^*) . It requires

$$n(p^*) = 1, \quad Y^* = O^*.$$

Replacement selects

$$p_0 = \frac{v_0 - q}{a} - \omega.$$

Housing clearing then selects $S_0 = S^*(p_0)$, where

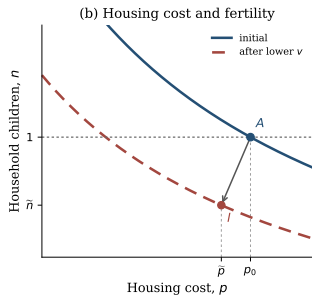
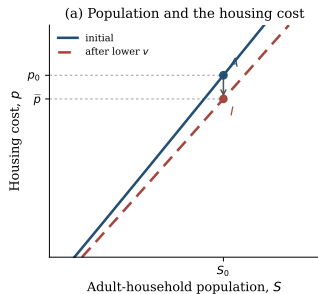
$$S^*(p) = \frac{2\bar{H}p^{1+\varepsilon}}{\alpha + \bar{h}_O + ap}.$$

Along the steady-state schedule, direct differentiation gives

$$\frac{dS^*}{dp} = \frac{2\bar{H}p^\varepsilon [(1 + \varepsilon)(\alpha + \bar{h}_O) + \varepsilon ap]}{(\alpha + \bar{h}_O + ap)^2} > 0 \quad \text{for } \varepsilon \geq 0.$$

Point A is the initial steady state.

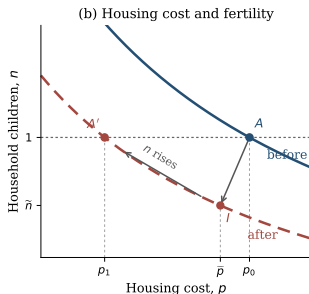
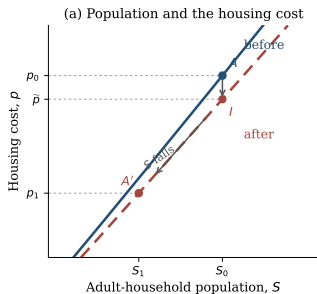
Impact response



- **Shock:** $v_0 \rightarrow v_1 < v_0$.
- **Inherited cohorts:** $Y_t = Y_0$,
 $O_t = O_0$, $S_t = S_0$.
- **Housing cost:** lower child-related demand gives
 $p_0 \rightarrow \tilde{p}$.
- **Fertility:** $\tilde{n} = n(\tilde{p}; v_1) < 1$.
- **Point I:** the impact equilibrium with inherited cohort sizes.

Demographic adjustment

► Dynamic path



- **Below replacement at l :**

$$Y_{t+1} = \tilde{n}Y_t < Y_t.$$

- **Cohorts:** smaller young cohorts eventually reduce S .
- **Housing cost:** lower demand moves $\tilde{p} \rightarrow p_1$.
- **Fertility:** lower p raises n along the red curve.
- **Point A' :** $n = 1$, $p_1 < p_0$, and

$$S_1 < S_0$$

Lower prices cushion the contraction if the fertility schedule crosses replacement at a positive population level.

Below-replacement fertility and population dynamics

Let $n_t = n(p_t)$ be household children per young household and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $n_t < 1$ whenever

$$\mu_t < n_t < 1.$$

A numerical example

$$100 \text{ young} + 70 \text{ old} \longrightarrow 80 \text{ young} + 100 \text{ old}$$

$$n_t = 0.8 < 1, \text{ but total households rise from 170 to 180.}$$

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1 - \mu_t)h_O(p_t) > (1 - n_t)h_Y(p_t).$$

Inherited cohorts can sustain short-run population and housing-cost growth below replacement.

Dated quantitative transition

Household problem. Households choose fertility, tenure, housing, saving, and location. The distribution G_t records their joint distribution and mass.

Calendar-time accounting.

$$G_{t+1} = \mathcal{T}(G_t, p_t) + E_{t+1}, \quad E_{t+5} = \frac{b_t^{\text{child}}}{2.1}.$$

The transition $\mathcal{T}$ moves households across age and economic states. Here b_t^{child} counts children; 2.1 children generate one future entrant household under the replacement normalization. Entrants arrive after twenty years.

Housing market. At each date, the housing cost clears demand from the current distribution:

$$H^s(p_t) = D(G_t, p_t).$$

- **2007–2023:** household totals and age shares are imposed; the model determines states within age.
- **After 2023:** cohort masses evolve under the demographic law above.

Calibration from 2007 to 2023

1. **Replacement normalization:** completed fertility equals 2.1.
2. **2007 state:** match age shares and preserve economic states conditional on age.
3. **2007–2023:** impose demographics; estimate eleven parameters from twelve moments.
4. **Validation:** compare untargeted fertility, timing, tenure, rooms, house prices, and rents.
5. **Post-2023:** hold preferences fixed; evolve the inherited distribution and entrant cohorts.

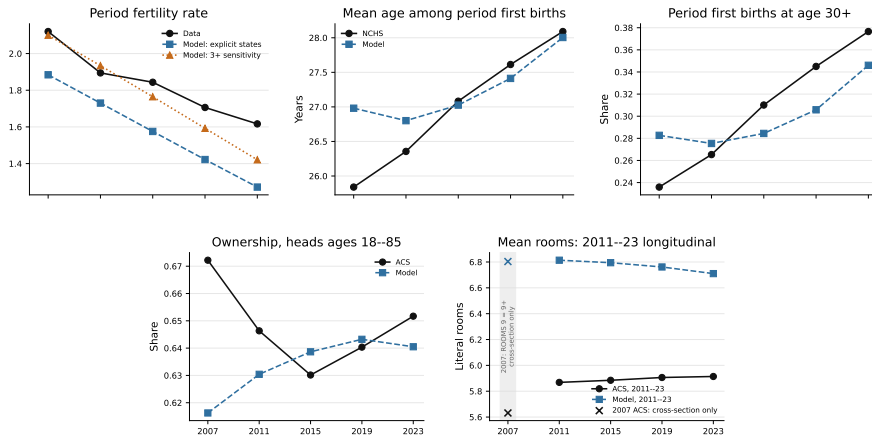
Demographic inputs: household totals and age shares. **Initial condition:** the 2007 distribution.

2023 calibration

Moment	Target	Model
Completed fertility	1.918	1.918
Childless share	0.188	0.186
Mean age at first birth	26.045	26.327
Share of first births at age 30+	0.260	0.242
Rooms response around first birth	0.720	0.380
Rooms gap: 3+ versus 1–2 children	0.368	0.359
Ownership gap: family versus no family	0.168	0.177
Ownership rate	0.575	0.546
Mean occupied rooms	5.780	6.710
Wealth / annual labor earnings	6.873	6.766
Annual bequests / aggregate wealth	0.0088	0.0086
Old-age wealth p90/p50	3.448	3.396

Largest discrepancies: the rooms response around first birth and mean occupied rooms.

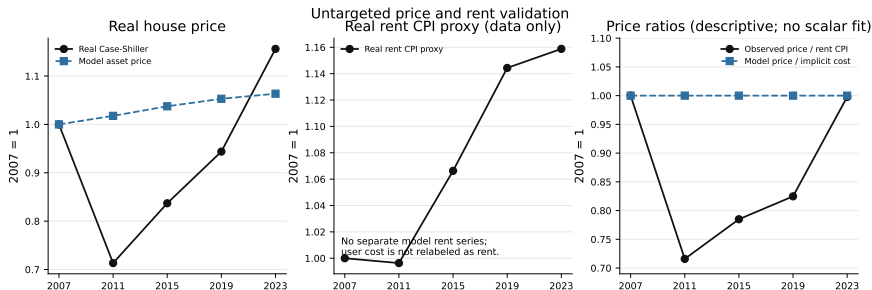
Historical validation



Birth timing is close by 2023; ownership and rooms dynamics are missed, and births per adult household fall too rapidly.

Sources: NCHS natality, World Bank period fertility, and ACS.

House-price dynamics



- The model misses the 2007–2011 house-price collapse and subsequent recovery.
- A fixed user-cost relation rules out independent price–rent dynamics.

The exercise isolates demographic-housing forces rather than the full housing cycle.

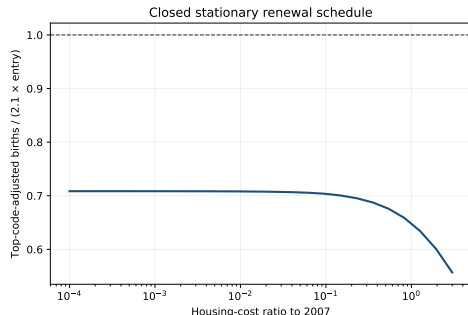
Sources: real Case-Shiller national house-price index and CPI rent of primary residence.

Options

Exercise	Exogenous demographic input	Endogenous outcomes	Interpretation
2007–2023 bridge	Household totals and age shares at each date	Prices and household choices within each age group	Historical accounting and 2023 calibration
Semi-open historical path	Four age-group residual flows	Cohort evolution around those flows	Requires a separate household-formation margin
Closed post-2023 path	The fitted 2023 state; no outside entry	Births, cohort sizes, prices, and tenure after 2023	National finite-horizon benchmark
Open post-2023 path	A fixed outside inflow and retention rate	The same endogenous choices and prices	Entry-flow sensitivity

Main specification: closed post-2023 path. **Sensitivity:** open post-2023 path.

Closed-population renewal



A positive closed steady state requires $B(P)/E(P) = 1$. Over the audited price grid,

$$0.557 \leq B(P)/E(P) \leq 0.708 < 1.$$

No price restores replacement. The closed continuation is therefore a finite-horizon contraction experiment, not a transition between two positive steady states.

Rebated property tax: capitalization

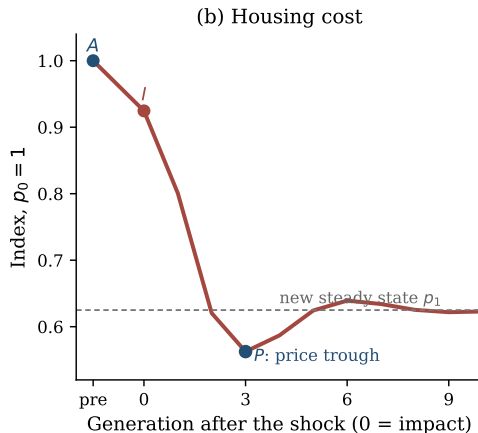
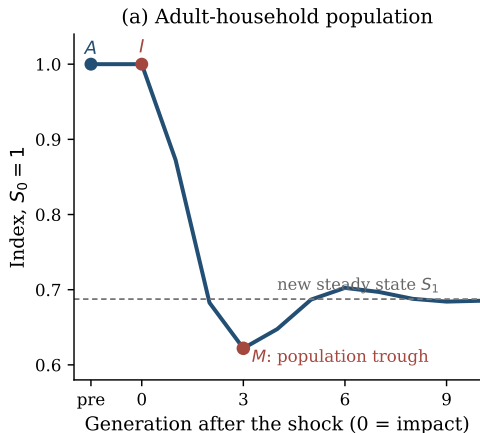
From the same fitted 2023 distribution, raise the annual property-tax rate from 1% to 2% and rebate the revenue equally. For each supply elasticity, re-anchor housing supply at the inherited price and solve jointly for the house price and transfer.

ε_H	Reference	ΔP	Δ user cost	Δ ownership	Δ child ownership	Δ births/adult
0.232	Coven: California	−8.02%	+14.18%	+1.80 pp	+2.53 pp	+0.52%
0.500	B–S–H: U.S. urban	−5.25%	+17.62%	+1.44 pp	+2.34 pp	+0.44%
1.750	Maintained national	−2.03%	+21.61%	+0.43 pp	+1.06 pp	+0.34%

A more elastic supply curve weakens price capitalization. With the maintained national elasticity, the rebate still raises ownership—especially among households with dependent children—and fertility rises modestly.

Housing supply: Baum-Snow and Han (2024), Saiz (2010), and the California calibration in Coven et al. (2025).

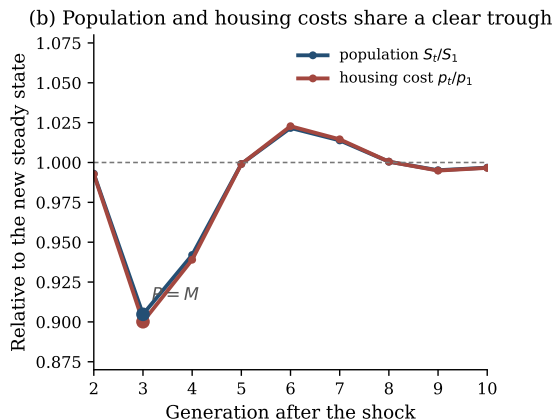
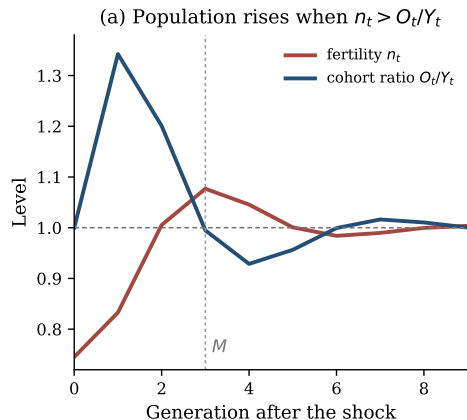
Population and housing-cost dynamics



Both series reach a trough in generation 3 before converging to lower steady states.
Illustrative parameterization.

◀ Equilibrium diagram

Cohort composition and the population trough



Population falls while $n_t < O_t/Y_t$, even after $n_t > 1$. Population and price troughs coincide here but need not when housing demand differs by age.